COMP 2711 – Mock Midterm 01

Solutions and marking notes

Practice material – not an official examination

Problem 1 [10 points]

(a) **[3 points]** By distribution, $(p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q) \equiv p \vee F \equiv p$. Negating gives $\boxed{\neg p}$. The steps use distributive, negation and identity laws.
Marking: 1 distribution; 1 contradiction/identity simplification; 1 final negation.

(b) **[2 points]** Invalid. Take $p = F, q = T$. Then $p \to q = T$ and $q = T$, but $p = F$. Thus all premises can be true while the conclusion is false.
Marking: 1 separating assignment; 1 verification of all premises and conclusion.

(c) **[5 points]**

1. $q \vee r$ from the first two premises by resolution.
2. $\neg q \vee s$ from $q \to s$ by implication equivalence.
3. $r \vee s$ from lines 1 and 2 by resolution.
4. $\neg r \vee s$ from $r \to s$ by equivalence.
5. $s \vee s$, hence s, from lines 3 and 4 by resolution and idempotence.
6. t from line 5 and $s \to t$ by modus ponens.

Marking: 1 first resolution; 1 implication conversion; 1 second resolution; 1 obtaining s; 1 final modus ponens with rule citations.

Problem 2 [13 points]

(a) **[4 points]** (i) $\forall x(S(x) \to \exists y(S(y) \wedge R(x,y)))$. (ii) $\exists y(S(y) \wedge \forall x(S(x) \to \neg R(x,y)))$. The reviewer is the first argument; the report owner is the second.
Marking: 2 each: 1 correct quantifiers/argument order; 1 correct student restrictions.

(b) **[4 points]**

$$\exists x\big(A(x) \wedge \forall y(\neg B(y) \vee \exists z \neg C(x,y,z))\big).$$

Negation switches each quantifier, turns the outer implication into a conjunction, and turns the inner conjunction into a disjunction.
Marking: 1 outer quantifier; 1 implication; 1 remaining quantifiers; 1 inner De Morgan step and scopes.

(c) **[5 points]**

1. Choose a fresh witness a with $P(a) \wedge S(a)$ by existential instantiation.
2. Obtain $P(a)$ and $S(a)$ by simplification.
3. Instantiate the two universal premises to $P(a) \to Q(a)$ and $Q(a) \to \neg R(a)$.

4. Obtain $Q(a)$ and then $\neg R(a)$ by modus ponens.
5. Conjoin $S(a) \wedge \neg R(a)$ and use existential generalization.

Marking: 1 fresh witness; 1 simplification; 1 universal instantiation; 1 implication chain; 1 conjunction/existential generalization.

Problem 3 [12 points]

(a) **[2 points]** For any x, $x \in (A \setminus B) \cap C$ iff $x \in A$, $x \notin B$, and $x \in C$; these are exactly the conditions for $x \in A \cap (C \setminus B)$. Thus the sets have the same elements.
Marking: 1 correct membership conditions; 1 both directions/equality.

(b) **[3 points]** If $2a + 1 = 2b + 1$, then $a = b$, so f is injective. Every $y \in O$ has the form $2k + 1$, so $f(k) = y$; hence f is onto. Its inverse is $f^{-1} : O \to \mathbb{Z}$, $f^{-1}(y) = (y - 1)/2$, which is an integer for all $y \in O$.
Marking: 1 injection; 1 surjection; 1 inverse and correct domains.

(c) **[7 points]**

(i) **T**. The singleton contains the empty set as its sole element.

(ii) **F**. Its element $\varnothing$ is not an element of the empty set.

(iii) **T**. A two-element set has 2^2 subsets.

(iv) **T**. Every listed pair belongs to the Cartesian product.

(v) **F**. The map $n \mapsto n + 1$ from $\mathbb{N}$ to $\mathbb{N}$ misses 0.

(vi) **T**. Order by length, then lexicographically; each length layer is finite.

(vii) **F**. It is in bijection with $\mathcal{P}(\mathbb{N})$ by the set of one-positions, and is uncountable.

Marking: Award 1 point for each correct T/F choice (7 total); reasons are included here for checking.

Problem 4 [16 points]

(a) **[2 points]** Choose the position of T, then fill the remaining positions: $3 \cdot 5 \cdot 4 = 60$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(b) **[2 points]** (i) Choose the common value and the two remaining values independently: $3^3 = 27$. (ii) 0, since four distinct inputs cannot have distinct images among three outputs.
Marking: 1 mark for each correct count.

(c) **[2 points]** The string is 10101. Each subset has a unique membership string and conversely each string gives a subset; five independent bits give $2^5 = 32$.
Marking: 1 mark for 10101; 1 mark for the bijective encoding and 2^5.

(d) **[2 points]** There are $3 \cdot 5$ output entries and four multiplications in each: $3 \cdot 5 \cdot 4 = 60$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(e) **[2 points]** Split by length and subtract the all-letter codes: $3(5-3)+3(5^2-3^2) = 54$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(f) **[2 points]** Compress the three-letter block. Its five units have 5! orders and its internal order has 3! choices, giving 5!3!.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(g) **[2 points]** Treat the adjacent pair as a circular block: $2(5-1)! = 2 \cdot 4!$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(h) **[2 points]** Choose four factors contributing $2x$: $\binom{7}{4}2^4(-1)^3 = -560$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

Problem 5 [10 points]

[10 points] Count triples (S, a, b) where S is an r-element subset of a fixed n-element set and $a, b \in S$ are distinct, ordered distinguished members. Choosing S, then a and b, gives $\binom{n}{r}r(r-1)$. Choosing the ordered pair first in $n(n-1)$ ways and then the other $r-2$ elements gives $n(n-1)\binom{n-2}{r-2}$. Every triple is produced exactly once by each procedure, including $r = 2$ and $r = n$.
Marking: 2 marks for the counted objects and ordered distinction; 3 for the first count; 3 for the second; 2 for explaining the exact correspondence.

Problem 6 [9 points]

[9 points] Put $u = x - 1$, $v = y$, $w = z - 2$. Then $u + v + w = 9$, with caps $4, 4, 5$. The unrestricted count is $\binom{11}{2}$. Let the bad events be $u \geq 5$, $v \geq 5$, $w \geq 6$. Their sizes are $\binom{6}{2}, \binom{6}{2}, \binom{5}{2}$ after shifting. Every pairwise intersection needs a total shift at least ten and is empty. Thus inclusion–exclusion gives

$$\binom{11}{2} - 2\binom{6}{2} - \binom{5}{2} = 15.$$

Marking: 2 marks for shifting lower bounds and defining the universe; 2 for bad events; 3 for singleton counts; 1 for empty overlaps; 1 for the final IE expression.

Problem 7 [7 points]

[**7 points**] For the sum, partition the integers into fifteen pairs $\{i, 31-i\}$ for $i = 1, \ldots, 15$. Sixteen selections force both members of one pair. For the difference, use the different partition $\{i, i+15\}$ for $i = 1, \ldots, 15$. Again sixteen selections force a complete pair. The fifteen-element selection $\{1, \ldots, 15\}$ contains neither a sum-31 pair nor a difference-15 pair, so sixteen is minimal.
Marking: 3 marks for the sum partition and PHP; 3 for the difference partition and PHP; 1 for the sharp selection.

Problem 8 [6 points]

(a) [**1 points**] S consists of the ordered triples of distinct labels; $|S| = 6 \cdot 5 \cdot 4 = 120$.
Marking: 1 mark for the ordered distinct triples and 120.

(b) [**1 points**] The specified triple is one outcome, so its probability is $1/120$.
Marking: 1 mark for $1/120$.

(c) [**1 points**] $P(E \cap F) = (3/6)(2/5) = 1/5$.
Marking: 1 mark for $1/5$.

(d) [**1 points**] $\binom{3}{3}/\binom{6}{3} = 1/20$.
Marking: 1 mark for $1/20$.

(e) [**2 points**] Both marginals are $1/2$, but $P(E \cap F) = 1/5 \neq 1/4$, so they are dependent. Their intersection has positive probability, so they are not disjoint.
Marking: 1 mark for the product comparison; 1 for the disjointness conclusion with reason.

Problem 9 [7 points]

(a) [**3 points**] Writing D for defect and A for alarm, the disjoint partition D, D^c gives $P(A) = (1/10)(4/5) + (9/10)(1/5) = 13/50$.
Marking: 1 mark for the partition; 1 for the weighted sum; 1 for $13/50$.

(b) [**4 points**] $P(D \mid A) = (4/50)/(13/50) = 4/13$. Also $P(A^c) = 37/50$ and $P(D \cap A^c) = (1/10)(1/5) = 1/50$, so $P(D \mid A^c) = 1/37$. Both conditioning events have positive mass.
Marking: 2 marks for each correct posterior with joint/conditioning denominator.

Problem 10 [10 points]

(a) **[2 points]** $E(D) = 1$ and $E(D^2) = 3/2$, so $V(D) = 3/2 - 1 = 1/2$.
Marking: 1 mark for the mean; 1 for the variance.

(b) **[2 points]** $E(Y) = 3 - 2 = 1$ and $V(Y) = (-2)^2(1/2) = 2$.
Marking: 1 mark for each result with correct coefficient handling.

(c) **[2 points]** Independence gives $E(DD') = E(D)E(D') = 1$, whereas $E(D^2) = 3/2$. Squaring the same variable is not an independent product.
Marking: 1 mark for each moment and the independence distinction.

(d) **[4 points]** Linearity gives $E(M) = 1$ without needing independence. Independence permits variance addition: $V(M) = (1/9)(3/2) = 1/6$. Thus $\sigma(M) = \sqrt{1/6}$. Scaling contributes $1/9$, not $1/3$.
Marking: 1 mark for the mean; 2 for independent variance addition and scaling; 1 for the standard deviation.